Trails — Product Requirements Document

v0.3.1 · Reliable Reference Frames and Retrace Alignment

Status: Corrective release
Platform: Native iOS
Depends on: v0.2.1 capture and persistence; v0.3.0 retrace implementation
Release focus: Make repeated walks comparable in the same physical reference frame

> Before Trails tells you where to go, it must know where you are relative to the original trail.

1. Problem statement

The current retrace experience produces substantially different paths when the user walks approximately the same route twice, particularly in small spaces.

The returning session does not reliably connect to the original physical reference.

Possible contributors include:

- Independent session origins and orientations.
- Missing or unsuccessful relocalisation.
- Tracking drift.
- Camera-session transitions or resets.
- Geographic uncertainty larger than the movement being measured.
- Incorrect coordinate conversion or rendering.

These are investigation hypotheses, not a confirmed diagnosis.

The release must identify the actual failure and fix it. Drawing the second path closer to the first without establishing physical alignment is not an acceptable solution.

2. Release objective

The central requirement is **alignment**.

A user can record a short indoor trail, return to the same area, establish the original reference frame, and see a live path that meaningfully corresponds to the original recording.

When that cannot be established, Trails explains the limitation and offers manual follow-along instead of pretending live positioning works.

Success means

- The original path remains unchanged.
- A new session has an explicit relationship to the original coordinate frame.
- Comparable paths use the same scale, orientation, and reference.
- Tracking loss produces a visible state change.
- Small-space performance is measured on real devices.
- Failed positioning never appears as successful retracing.

3. Scope

Required

- Audit the current position source and coordinate pipeline.
- Separate geographic and local spatial coordinates.
- Persist reference-frame metadata.
- Implement explicit indoor alignment states.
- Validate saved-map relocalisation on supported devices.
- Preserve tracking segments across interruptions.
- Fix coordinate/rendering inconsistencies.
- Add positioning diagnostics.
- Add manual follow-along fallback.
- Validate short-route repeatability with physical reference points.

Optional controlled-demo capability

- Marker-based alignment using a known physical reference, if automatic relocalisation is not sufficiently reliable.

This must be labelled as a distinct mode, not presented as markerless positioning.

Not included

- Turn-by-turn navigation.
- AI-generated directions.
- OSM routing.
- Automatic snapping of live positions onto the recorded trail.
- Automatic merging of independent spatial captures.

- Guaranteed indoor positioning everywhere.
- Public sharing.

Existing capture and media features must continue working.

4. Release strategy

Implement the correction in this order:

```
Understand current measurements
→ Verify coordinate conversion and rendering
→ Measure within-session tracking
→ Establish cross-session alignment
→ Handle interruptions
→ Re-enable retrace feedback
```

Do not start by adding more navigation UI.

Diagnostic decision table

Observation	Investigation priority
A stationary device appears to move substantially	Measurement stability
A loop fails to return near its starting position within one session	Tracking quality and drift
Individual recordings look reasonable but do not overlap	Cross-session alignment
Failure begins after entering/exiting camera or scan modes	Session lifecycle and resets
Stored positions agree but drawings differ	Rendering and coordinate conversion

5. Positioning modes

Every trail and retrace session must declare its positioning mode.

Mode	Intended experience
Geographic	Outdoor path comparison using geographic observations
Local spatial, aligned	Live indoor retrace in an established shared frame
Local spatial, unaligned	Independent recording; no shared-position claim
Manual follow-along	Original moments/plan with user-controlled progression

Do not mix these modes implicitly.

A geographic observation must not be treated as a precise indoor position simply because it exists.

A room model must not be treated as proof that the live device has been positioned inside it.

6. Reference-frame contract

Every local path belongs to a named reference frame.

Required metadata:

```
ReferenceFrame
  id
  sourceSessionId
  units
  axisConvention
  worldAlignmentConfiguration
  createdAt
  positioningMethod
  alignmentAssetReferences[]
```

Coordinate rules

- Store original observations without overwriting them.
- Record which frame owns every local position.
- Use explicit transforms when converting between frames.
- Document transform direction and matrix convention.
- Apply transforms consistently to paths, moments, and live pose.
- Never combine coordinates solely because both are measured in metres.
- Never infer a shared frame from matching trail or place names.

Rendering rules

- Both paths use one shared rendering coordinate system.
- Preserve metric scale.
- Do not independently rotate or normalise paths before comparison.
- Camera fitting may change the viewport, not the underlying geometry.
- Display tracking gaps as gaps.

7. Original-trail recording requirements

Future recordings intended for indoor retracing must preserve:

- Local camera/path observations.
- Observation timestamps.
- Tracking state.
- Session-segment identifiers.
- Coordinate-frame identifier.

- Available relocalisation assets.
- The capture configuration needed to interpret them.
- A reference-area image, where captured and permitted.
- Optional user-provided starting-area description.

Reference-area UX

Suggested guidance:

> "For easier retracing, start near a recognisable area and look around before walking."

At finish, determine retrace capability from the assets actually saved.

Possible outcomes:

- Indoor alignment available.
- Manual follow-along only.
- Spatial asset processing incomplete.

Do not mark an old trail as alignment-capable merely because it contains a plan or 3D model.

8. Indoor retrace state machine

```
Unavailable
or
Ready to attempt alignment
  → Loading reference
  → Searching
  → Verifying
  → Aligned
  → Tracking degraded
  → Alignment lost
  → Searching again
```

State requirements

State	User experience
Unavailable	Explain missing reference data; offer manual follow-along
Loading	Load assets; do not show a live aligned marker
Searching	Guide user to look around the reference area
Verifying	Check the restored frame against the reference
Aligned	Show live position and retrace path
Degraded	Reduce confidence; pause unreliable metrics
Lost	Stop presenting the live marker as trustworthy; offer recovery

A generic “tracking normal” callback alone must not be the application’s proof of restored alignment.

The implementation must document the evidence used to transition into Aligned.

9. Alignment verification

Automatic alignment

Implementation requirements:

1. Load the saved reference assets.
2. Attempt relocalisation against that reference.
3. Track the restoration lifecycle explicitly.
4. Check pose stability over an observation interval.
5. Validate against available reference anchors or checkpoints.
6. Enable aligned rendering only after the verification criteria pass.

User confirmation

A user may confirm that a displayed reference appears correct, but confirmation must not silently substitute for missing technical alignment.

Manual start-point selection

“I am at the start” is a location assertion, not a full pose measurement.

Do not use a single tap to fabricate both translation and orientation.

If manual alignment is offered:

- Define the additional direction/reference input.
- Label the resulting mode accurately.
- Do not claim the same confidence as validated automatic alignment.

10. Controlled marker-based mode

If needed for an honest, repeatable demonstration, support an explicit reference-marker workflow.

Flow

```
Record using a known reference marker
→ Save marker relationship and path
→ Return
→ Observe the same marker
→ Verify alignment
→ Retrace
```

Requirements:

- Known physical marker dimensions.
- Stable placement.
- Persisted marker identity and relationship to the trail.
- Clear indication that the marker must not move.
- Recovery when the marker cannot be recognised.
- Separate measurements of alignment quality and subsequent tracking drift.

This mode is optional. It must not delay fixing a simpler coordinate or rendering bug.

11. Tracking interruptions and camera ownership

Audit all transitions between:

- Photo/video capture.
- Indoor scanning.
- Retrace tracking.
- Background/foreground.
- Interrupted/resumed sessions.

Required behaviour

- One coordinator owns camera/session transitions.
- Every tracking reset creates an explicit segment boundary.
- No automatic straight line connects incompatible segments.
- Alignment is re-evaluated after a reset.
- Previously saved media remains accessible.

If reliable simultaneous camera and spatial tracking is not supported by the current implementation, use a deliberate transition.

A seamless-looking screen is less important than preserving coordinate truth.

12. Live retrace interface

Aligned state

Show:

- Original trail: dashed purple.
- Current walk: solid teal.
- Current position.
- Selected moments.
- Alignment status.
- Recenter control.

Use line styles and labels in addition to colour.

Not yet aligned

Show the original reference without a misleading moving marker.

Suggested copy:

> “Finding your position in this trail.”

Supporting instruction:

> “Return near the starting area and slowly look around.”

Lost alignment

Suggested copy:

> “Positioning lost. Your recorded trail is still available.”

Actions:

- Try alignment again.
- Follow manually.
- End retrace.

A last-known marker may remain only if clearly labelled as stale.

13. Metrics and confidence

Enable metrics only when their input positioning is valid.

Allowed after validation

- Distance travelled in the current continuous segment.
- Approximate distance from the original path.
- Estimated progress along the original path.
- Approximate distance to the next positioned moment.

Required safeguards

- Mark estimates as approximate.
- Pause or hide metrics when positioning is degraded.
- Handle loops and path crossings.
- Do not select progress solely from the nearest point.
- Do not silently reset progress after alignment loss.
- Do not display centimetre-level precision unsupported by measurements.

Off-route alerts remain disabled until alignment and path-matching behaviour pass validation.

14. Manual follow-along fallback

This is a valid product mode, not an error screen.

Available features:

- Original trail.
- Saved plan, if present.
- Ordered moments.
- Photos, video, voice, and notes.
- “I’m here” selection.
- Previous/Next moment.

Clearly state:

> “Manual follow-along — live position is unavailable.”

Do not generate a simulated live path from manually selected moments.

15. Data-model additions

```
RetraceSession
  id
  sourceTrailId
  startedAt
  endedAt?
  positioningMode
```

```
  alignmentState
  alignmentMethod?
  referenceFrameId?
  segmentIds[]
  visitedMomentIds[]
```

```
TrackingSegment
  id
  sessionId
  coordinateFrameId
  startedAt
  endedAt?
  rawPathReference
  transformToReference?
  transformStatus
  interruptionReason?
```

```
AlignmentAttempt
  id
  retraceSessionId
  referenceAssetIds[]
  method
  startedAt
  endedAt?
  outcome
  verificationResults
  failureReason?
```

Existing trails must migrate without fabricated reference assets or transforms.

16. Diagnostics

Provide a developer-only positioning panel showing:

- Position source.
- Current session and segment IDs.
- Reference-frame ID.
- Reference assets loaded.
- Alignment lifecycle.
- Tracking state.
- Raw pose.
- Transformed pose.
- Active transform.
- Last interruption/reset.
- Geographic accuracy when applicable.

Provide a local diagnostic export containing configuration, state transitions, and positioning samples.

Do not include photos, audio, or video by default.

The goal is to distinguish a measurement failure from an alignment failure or drawing bug.

17. Validation plan

Test on declared supported devices using marked physical checkpoints.

Required scenarios

1. Stationary observation.
2. Short straight walk.
3. Small loop returning to the start.
4. Repeated route within one session.
5. Repeated route after application restart.
6. Different phone orientation at the starting area.
7. Camera-to-scan transition.
8. Tracking interruption and recovery.
9. Low-detail or changed reference area.
10. Missing or invalid saved alignment assets.

Measure

- Starting alignment error.
- Error at physical checkpoints.
- Loop-closure error.
- Time to alignment.
- Alignment success rate.
- False-success rate.
- Behaviour after tracking loss.

Do not judge correctness solely by how closely two rendered lines overlap.

18. Proposed release gates

These are initial engineering targets, not claims about current capability. Validate or tighten them before release; do not quietly relax them to conceal failures.

Controlled short indoor route

For a marked 5–15 metre route:

- At least 9 of 10 alignment attempts succeed within 30 seconds in the declared supported setup.
- Start/checkpoint error is within 0.5 metres in at least 9 of 10 measured trials.
- No deliberately unaligned trial is presented as successfully aligned.
- Tracking interruptions always produce the expected degraded/lost state.
- Original recordings remain unchanged.

Suitability boundary

Even meeting a 0.5-metre target does not qualify the app to distinguish every adjacent doorway or narrow passage.

The release must not claim a precision finer than its validation supports.

If the target is not achieved, ship manual follow-along and restrict automatic indoor retrace to configurations that pass.

19. Implementation priority

P0 — Find the actual defect

- Audit coordinate source.
- Inspect raw observations.
- Verify units, axes, transforms, and drawing.
- Reproduce with physical checkpoints.

P0 — Establish shared frames

- Save reference metadata/assets.
- Implement alignment lifecycle.
- Gate live overlay on verification.

P0 — Handle failure honestly

- Segment resets.
- Degraded/lost UI.
- Manual fallback.
- No fabricated snapping.

P1 — Validate the experience

- Device matrix.
- Repeatability tests.
- Recovery tests.
- Metric confidence rules.

P2 — Improve convenience

- Reference-area guidance.
- Controlled marker mode.
- Faster recovery.
- Additional diagnostic tooling.

20. Definition of done

Trails v0.3.1 is complete when:

> A returning user can establish the original reference, walk a short route again, and see a physically meaningful comparison—or receive a clear explanation that live alignment is unavailable.

The fix is not making two lines look similar.

It is making both lines mean positions in the same place.